

Kishoreganj University
2nd Year 2nd Semester B.Sc. (Engg.) Final Examination-2023
Department of Computer Science and Engineering
CSE 2205: Theory of Computing (3 Credits)

Time: 3 Hours

Full Marks: 70

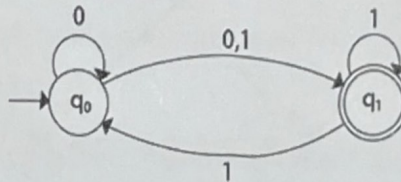
Instructions

- Figures shown in the right margin indicate full marks.
- Answer 05 out of 07 questions.

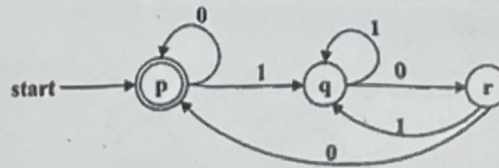
a. Define a dead state in automata. Why do we add these states in an automaton? 4

b. Draw a non-deterministic finite automaton that accepts 00 and 11 at the end of a string containing 0, 1 in it; e.g., it accepts 01010100 and 01010111, but not 000111010. 4

c. Convert the given NFA into its equivalent DFA. 6



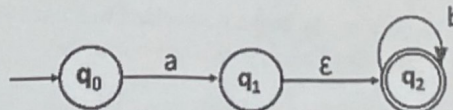
a. Construct a regular expression corresponding to the state diagram given in the following figure: 4



b. Define ϵ -closure. Compute ϵ -closure of each state and find its equivalent DFA. 6

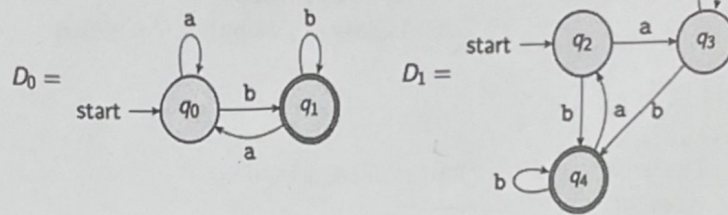
	ϵ	a	b	c
$\rightarrow p$	{q}	{p}	$\varnothing$	$\varnothing$
q	{r}	$\varnothing$	{q}	$\varnothing$
*r	$\varnothing$	$\varnothing$	$\varnothing$	{r}

c. Convert the following NFA with ϵ to NFA without ϵ . 4

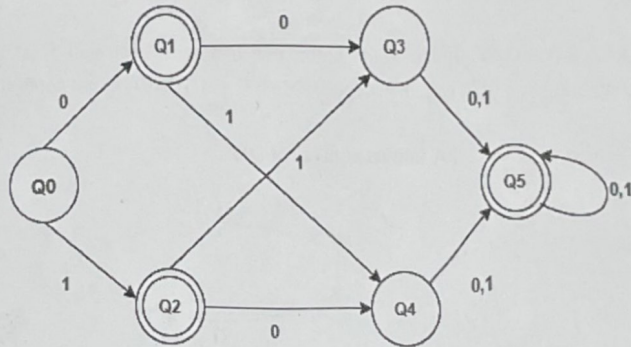


a. Explain the concept of equivalence in automata. What is the difference between equivalence of automata and equivalence of languages? 4

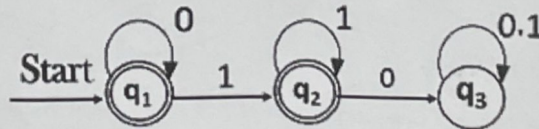
- b. Are the following two DFA equivalent?



- c. Minimize the following DFA using Myhill-Nerode Theorem.



- 4 a. Design a finite automaton from the regular expression $10 + (0 + 11)0^*1$.
- b. "A language L is accepted by some DFA if and only if L is accepted by some NFA"-Prove this statement.
- c. Construct the regular expression for the given DFA using Arden's Theorem.



Mealy and Moore machines were developed to model sequential logic systems, providing formal methods to describe output behavior based on states and inputs. These machines offered solutions for different design needs: Moore machines for more predictable, state-dependent outputs and Mealy machines for more dynamic, input-responsive outputs.

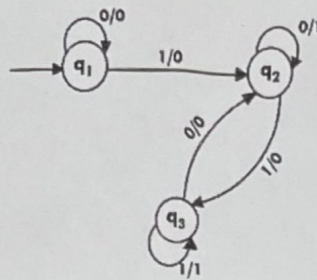
In the Moore machine, the output is associated with every state, and in the Mealy machine, the output is given along the edge with an input symbol.

Now, answer the following questions.

- a. Design a Moore machine to generate 1's complement of a given binary number.

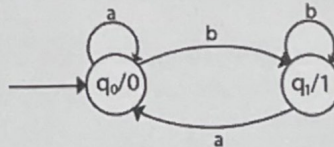
- b. Convert the following Mealy machine into an equivalent Moore machine.

5



- c. Convert the following Moore machine into its equivalent Mealy machine.

5



- 6 a. Define Pushdown Automata with formal notations. Draw a PDA for the language $L_{ww^R} = \{WW^R \mid W \text{ is in } (0+1)^*\}$.

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- b. How does a Pushdown Automaton (PDA) differ from a Finite Automaton (FA)?

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- c. Convert the following grammar to a PDA that accepts the same language by empty stack.

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$$S \rightarrow 0S1 \mid A$$

$$A \rightarrow 1A0 \mid S \mid \epsilon$$

Simulate the action for the input 0011 on the new PDA.

- 7 a. Design a Turing Machine (TM) to implement the function "multiplication". Simulate the action for the input 001000.

4

- b. What is the distinction between undecidability and intractability in the context of automata theory? Why is the exploration of these concepts important for understanding the boundaries of computational problems?

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- c. Discuss how the concept of undecidability contributes to the field of cryptography. Provide necessary examples.

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